Habib University

shaping futures

Course Title: **LINEAR ALGEBRA**
Instructors:
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Class ID: **MATH 205**
Total Marks**: 100**
Duration**: 3 hours**

	Marks Obtained
Q. 1	/12
Q. 2	/8
Q. 3	/6
Q. 4	/10
Q. 5	/10
Q. 6	/12
Q. 7	/10
Q. 8	/12
Q. 9	/10
Q. 10	/10
Total	/100
Instructor's Signature & Date	

Instructions for Students:

a. Read the following instructions carefully.
b. Please ensure that your mobile phone is switched off/on silent, and left in your bag or at the front desk in the examination room. **You must not have a phone on your person during the examination.**
c. **All your personal belongings must be placed at the designated area of the examination hall.**
d. Write your Name, ID, Section, Instructor's name, and date in the space provided below.
e. Remember to sign the exam attendance sheet circulated by the head invigilator/proctor.
f. Answer **all** questions in the exam.
g. You must provide complete working and reasoning for all steps of your solution.
h. Try to complete working in the space provided below each question. In the case that you run out of space, use the blank pages provided to continue your working, labelling your work clearly.
i. If you need to leave the room for any reason, please raise your hand and take permission from the head invigilator/proctor.
j. After finishing your exam, raise your hand and give your answer book to the head invigilator/proctor.

Name: ____

Student ID: ___________________________ Section: ________________

Instructor: ____

Date: ____

Question 1 **[12 marks]**

Prove equivalence of the following statements in the order:

$$\text{(a)} \Longrightarrow \text{(b)}, \text{(b)} \Longrightarrow \text{(c)}, \text{(c)} \Longrightarrow \text{(d)}, \text{(d)} \Longrightarrow \text{(a)}$$

If A is an n x n matrix, then the following statements are all true or all false:

(a) $A\boldsymbol{x} = \mathbf{0}$ has only the trivial solution.
(b) The reduced row-echelon form of A is I_n.
(c) A is expressible as a product of elemenatry matrices.
(d) A is non-singular.

Question 2

Prove: Let V be a finite-dimensional vector space, and let $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \dots, \boldsymbol{v_n}\}$ be any basis. If a set has more than n vectors, then it is linearly dependent. **[8]**

Question 3 Let S be the set of all real numbers in $(0, \infty)$. If S is a vector space under the operations $\underline{\boldsymbol{u}} + \underline{\boldsymbol{v}} = \underline{\boldsymbol{u}}\,\underline{\boldsymbol{v}}$ and $\boldsymbol{k}\underline{\boldsymbol{u}} = \underline{\boldsymbol{u}}^{\boldsymbol{k}}$, then identify the additive identity $\underline{0}$ of S and the additive inverse $-\underline{u}$. Hence, show that Axioms 4 and 5 for vector spaces are satisfied for S. **[6]**

Question 4 [10 marks]

If V is an $n-$dimensional vector space, and if S is a set in V with exactly n vectors, then S is a basis for V if either S spans V or S is linearly independent.

Question 5: [10 Marks]

Prove: If W is a set of one or more vectors from a vector space V, then W is a subspace of V **if and only if** the following conditions hold:

(a) If $\mathbf{u}$ and $\mathbf{v}$ are vectors in W, then $\mathbf{u} + \mathbf{v}$ is in W.
(b) If k is any scalar and $\mathbf{u}$ is any vector in W, then $k\,\mathbf{u} \in W$.

Question 6 [12 marks]

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the matrix transformation $T(\boldsymbol{x}) = A(\boldsymbol{x})$, where T is the linear operator defined by the formula

$$T(x_1, x_2, x_3) = (x_1 + x_2 + 2x_3,\ x_1 + \ x_3, 2x_1 + x_2 + \ 3x_3).$$

Find the bases for the kernel and range of T. Hence, state the dimension of ker (T) and R (T).

<u>Question 7</u> [10 marks]

Prove: A rotation matrix is a transition matrix from one orthonormal basis S to another orthonormal basis T.

Question 8 **[6] + [6]**

(a) If k is a positive integer, λ is an eigenvalue of matrix A, and X is a corresponding eigenvector, then prove that λ^k is an eigenvalue of A^k and X is a corresponding eigenvector.

(b) Give a general proof of the Cauchy-Schwarz Inequality: $|\langle \boldsymbol{u}, \boldsymbol{v} \rangle| \leq \|\boldsymbol{u}\| \, \|\boldsymbol{v}\|$, if $\boldsymbol{u}, \boldsymbol{v} \in V$.

Question 9 (a)

In each part, use the information in the table to find the dimension of the row space, column space, and null space of A. **[6]**

	Part (i)	Part (ii)
$\boldsymbol{Size\ of\ A}$	5×9	$6\ \times 2$
$\boldsymbol{Rank(A)}$	2	2

Question 9 (b)

Check whether the following vectors form a basis for $\boldsymbol{P_2}$? **[4]**

$$\{4 + 6x + x^2, -1 + 4x + 2x^2, 5 + 2x - x^2\}$$

Question 10 [10 marks]

Consider matrix $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$, with the details as per the table on the right. Find an orthogonal matrix P that diagonalizes A. You must show that P in fact does produce a diagonal matrix from A.

Eigenvalues	**Eigenvectors**
$\lambda_1 = \lambda_2 = 2$	$\boldsymbol{u_1} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \boldsymbol{u_2} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$
$\lambda_3 = 8$	$\boldsymbol{u_3} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

ROUGH WORK